UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

FRIDAY, 10 MAY 2024 TIME : 2.30 PM – 4.30 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATION AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B.**
Each question carries 25 marks.

Candidates are allowed to use non-scientific calculator.

Answer questions only in the answer booklet provided.

SECTION A

Q1. (a) Propose an Amazon Web Services (AWS) architecture diagram in scaling your cloud infrastructure. Your design should consider:

- Choose an Amazon Machine Image (AMI) for your running web instance(s).
- Setup relational database(s) in storing web data.
- Perform application load balancing.
- Create an auto scaling group between availability zones.
- Automatically scale new instances within a private subnet.
- Allow your cloud resources in the private subnet in accessing internet.
- Create alarms and monitor performance of your infrastructure, and
- Integrating some relevant intelligent cloud services.

(10 marks)

(b) There are different sensors installed at various locations of an industrial plant, continuously gathering data from machines and devices. The data is crucial to be analyzed in real-time with a proper analytics tool, so that faults and failures can be resolved in minimum time. This is the core purpose of an industrial IoT system. Based on this scenario, which cloud deployment model would you recommend to the plant owner? Justify your selection with **TWO (2)** supporting reasons. (4 marks)

(c) You have an Amazon Elastic Compute Cloud (EC2) Instance in development that interacts with the AWS DynamoDB. You will deploy the EC2 Instance in production environment soon. Which AWS service(s) should be used for a secure communication between the EC2 Instance and the DynamoDB? Justify your answer. (3 marks)

(d) Generate a smart car solution (architecture diagram) using the AWS cloud platform. Your design should include the collection of various data from the car, storing the data in cloud storage, and utilizing a database for further analysis and visualization. (8 marks)

[Total : 25 marks]

Q2. (a) From your point of view, how does ChatGPT utilize cloud computing technology to enhance its natural language processing capabilities? Please provide **THREE (3)** points to support your justification. (6 marks)

(b) With so many different vendors producing IoT devices, such as sensors, there has been a lack of adequate focus from cybersecurity professionals on IoT's security issues. For the time being, propose **TWO (2)** temporary solutions in solving potential security loopholes in IoT. (6 marks)

Q2. (Continued)

(c) What do you understand about middleware? Explain **TWO (2)** benefits of middleware for distributed system programmers, providing an example for each benefit. (5 marks)

(d) Compare and contrast the cluster computing architecture with the peer-to-peer architecture in the context of distributed systems. Discuss the key differences in their **design**, **functionalities**, **security**, and **scalability**, and present the information in a table. (8 marks)

[Total : 25 marks]

Q3. (a) Suppose there are four processes: A, B, C, and D. All clocks run at the same rate (2 units of time), but initially, A's clock reads 78, B's clock reads 88, C's clock reads 44, and D's clock reads 66. At time 78 on A's clock, A sends a message to B. This message takes 6 units of time to reach B. Subsequently, B waits for 4 units of time and then sends a message to C, which takes 2 units of time to reach C. Later, C waits for 6 units of time before sending a message to D. The message reaches D after 2 units of time. After D waits for 4 units of time, another message takes 2 units of time to reach B again. Assuming that the system implements Lamport's timestamps, draw a picture to illustrate the timestamps for the messages and how they are obtained. (8 marks)

(b) Given a scenario of **4 Generals *AND* 1 Traitor**, find the traitor by using Byzantine Generals algorithm. Show all the steps and diagram necessary to your answer. (10 marks)

(c) Processes W, X, Y, and Z are executing at 4 distributed sites. Suppose the timestamps assigned to them (at the time of their creation) are 23, 77, 45, and 66, respectively. Y acquires a shared resource. How can a deadlock be prevented in the system if it implements the Wait-Die (WD) scheme and the Wound-Wait (WW) scheme when W requests the same resource held by Y? (4 marks)

(d) A clock, *X*, is reading 20:14:40.0 (hr:min:sec) when a clock synchronization request is sent to time server, *S*. After receiving the request from *X*, *S* prepares a response and appends the time 20:14:58.0 from its own clock. What will be the time set by *X* if the round-trip time (RTT) takes 8.0 seconds? Show your formula and calculation using Cristian's algorithm. (3 marks)

[Total : 25 marks]

SECTION B

Q4. (a) Explain **THREE (3)** key integration methods available for building distributed systems. (6 marks)

(b) A cloud application requires a database service. Do you believe that web services such as Amazon Simple Storage Service (S3), Google Storage, and Windows Azure Storage are suitable for hosting a database in the cloud? Justify your answer. (5 marks)

(c) Provide **THREE (3)** characteristics of cloud architecture that separate it from the traditional one. (6 marks)

(d) Answer the following questions with short justification in the context of AWS.

(i) You want to add an extra layer of protection on top of your username and password. Which AWS service can help in this regard? (2 marks)

(ii) UTAR is planning to host a large distributed application on the AWS platform. One of their major security concerns is DDoS attack. Which AWS cloud services can help alleviate this concern? (2 marks)

(iii) What acts as a firewall in AWS that controls the traffic allowed to reach one or more instances? (2 marks)

(iv) What is the best way to back up an Amazon Elastic Block Store (EBS) volume in AWS? (2 marks)

[Total : 25 marks]

Q5. (a) Discuss **THREE (3)** primary challenges in managing and maintaining a cluster computing system, and how are these challenges addressed? (9 marks)

(b) Recommend **ONE (1)** type of distributed computer system which is suitable to support 8 classes of students (40 students per class), where different classes required different types of mathematic/computer applications. Suppose that UTAR can only hire one lab administrator to take care of the machines. Provide **TWO (2)** justifications for your selection of distributed computer system. (3 marks)

(c) You hosted an application on AWS that lets users to render images and do some general computing. Which of the below listed services can be used to route the incoming user traffic? Justify your answer.

Q5. (c) (Continued)

- Classic Load Balancer
- Application Load Balancer
- Network Load balancer (4 marks)

(d) Multi-Agent System (MAS) has shown significant impact in solving large scale distributed systems for the past few decades.

(i) How do you define non-player characters with Artificial Intelligence (AI) in the multiplayer online battle as a distributed system? (4 marks)

Field	Value
Purpose	INFORM
Sender	max@http://fanclub-beatrix.royalty-spotters.nl:7239
Receiver	elke@iiop://royalty-watcher.uk:5623
Language	Prolog
Ontology	genealogy
Content	female(beatrix),parent(beatrix,juliana,bernhard)

Figure 5.0

(ii) Figure 5.0 shows an example of agent communication message using FIPA ACL format. Interpret this message. (3 marks)

(iii) Besides multiplayer online games, name **TWO (2)** more MAS applications in the real world. (2 marks)

[Total : 25 marks]